1. Please fill the answers of the following questions in the answer sheet given below: (68%)

(1) Let $x(t) \xleftrightarrow{\mathcal{F}} X(f)$ denotes the Fourier transformation of $x(t)$. Given $y(t) = x(t)e^{j8\pi t}$, then $Y(f) =$ $\underline{\quad X(f-4) \quad}$

(2) Let $y(t) = \int_{-\infty}^{\infty} x_1(t-\lambda)x_2(\lambda)d\lambda$, then $y(t) \xleftrightarrow{\mathcal{F}} Y(f) =$ $\underline{\quad X_1(f)X_2(f) \quad}$

(3) $x(t) = 2 \xleftrightarrow{\mathcal{F}} X(f) =$ $\underline{\quad 2\delta(f) \quad}$

(4) For all $m, n \in \mathbb{Z}$, find $\int_{-\infty}^{\infty} \operatorname{sinc}(100t-m)\operatorname{sinc}(100t-n)\,dt =$ $\underline{\quad \frac{\delta_{m,n}}{100} \quad}$

(5) For a frequency selective channel of response $\{h_n\}_{n=0,\cdots,5}$, let x_n and y_n denoting the input and output signals of the channel, respectively. With an AWGN noise z_n, please show $y_n =$ $\underline{\quad \sum_{l=0}^{5} h_l x_{n-l} + z_n \quad}$

(6) In (5), assuming that $\{h_n\}$ is known at the transmitter and d_n is the data symbol at time n, for $n \geq 6$, please show the output of the precoder, $x_n =$ $\underline{\quad \frac{d_n}{h_0} - \sum_{l=1}^{5} x_{n-l}\frac{h_l}{h_0} \quad}$

(7) In (5), if an OFDM system of N subcarriers is employed to deal with the ISI, and X_k is the data symbol delivered by the k^{th} subcarrier, please show the transmitted signal in the time domain, $x_n =$ $\underline{\quad [7]\ \frac{1}{\sqrt{N}}\sum_{k=0}^{N-1} X_k e^{\frac{2\pi}{N}kn} \quad}$

(8) In (7), let $H_k = \frac{1}{\sqrt{N}}\sum_{n=0}^{5} h_n e^{-j\frac{2\pi}{N}kn}$, then in the absence of noise please show the received signal at the k^{th} subcarrier in the frequency domain, $Y_k =$ $\underline{\quad [8]\ \sqrt{N}H_k X_k \quad}$

(9) Given a circularly symmetric Gaussian random variable $X = X_{\mathrm{R}} + jX_{\mathrm{I}}, \sim \mathcal{CN}(0,1)$, find $E\{X_{\mathrm{R}}X_{\mathrm{I}}\} =$ $\underline{\quad 0 \quad}$

(10). Consider a MIMO system employing $M = 8$ transmit and $N = 10$ receive antennas, the information is modulated by a QPSK signal, please calculate the spectral efficiencies, in bits/s/Hz, of VBLAST system. Ans: $\underline{\quad 16 \quad}$

(11) In (10), find the spectral efficiencies of SM system. Ans: $\underline{\quad 5 \quad}$

(12) In (10) and (11), let $\mathbf{s} = [s_1, \ \cdots, \ s_M]^T$ denotes the transmitted signal vector of the VBLAST system. Given $E\{|s_m|^2\} = 1.5, \forall m,$ please find the transmitted signal power of the active antenna of the SM system, so that the two system are of equal power consumption $E\{|s_{active}|^2\} =$ $\underline{\quad 12 \quad}$

(13) Please find the function of the MATLAB code given below.

Ans:? pseudo-spectrum of the MUSIC algorithm

(14) In (13), find the noise power, in dB, $E\left\{\|\text{noise}\|^2\right\} =$ $10\log_{10}(0.2)$

(15) In (13), there is a bug in the code sequences, find it out. Ans: Vn is incorrect

(16) In (13), determine the size of the variable D = 20×20

```
%==================================================================
x=[12, 18, -21];
sn=100;
K=length(x);
A=exp(-j*pi*[0:19]'*sin(x/180*pi));
noamp=0.1;
r=[];
for t=1:No_sn,
    noise= noamp/sqrt(2)*(randn(20,1)+i*randn(20,1));
    s_t=1/sqrt(2)*(randn(K,1)+j*randn(K,1));
    r_t=A*s_t+noise;
    r=[r,r_t] ;
end
Rr=r*r'/No_snapshots;
[Vn,D]=eig(Rr);
theta=[-90:.01: 89.99];
for k=1:length(theta),
    a_k=exp(-j*(0:9)'*pi*sin(theta(k)*pi/180));
    Spectra(k)=1/abs(a_k'*Vn*Vn'*a_k);
end
%==================================================================
```

In Q2-Q3, please provide detailed steps of calculations/ derivations.

2. Given a complex variable $z = x + jy$, the derivative operator is defined by $\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - j\frac{\partial}{\partial y}\right)$, please calculate the following with detailed derivations (12%)

(a) $\frac{\partial}{\partial z} z^2 z^* = ?$

(b) For a vector of complex variables $\mathbf{z} = \begin{bmatrix} z_1 \\ \vdots \\ z_3 \end{bmatrix}_{3\times 1}$ the derivative is defined by $\frac{\partial}{\partial \mathbf{z}} = \begin{bmatrix} \frac{\partial}{\partial z_1} \\ \vdots \\ \frac{\partial}{\partial z_3} \end{bmatrix}$,

then given a constant vector $\mathbf{p} = \begin{bmatrix} 1+2i \\ 2+4i \\ 3 \end{bmatrix}$, calculate $\frac{\partial}{\partial \mathbf{z}} \mathbf{z}^T \mathbf{p} = ?$

⟨Sol.⟩

(a)

• $z^2 z^* = \left(x^2 - y^2 + j2xy\right)\left(x - jy\right) = \left(x^3 + xy^2\right) + j\left(y^3 + x^2 y\right)$

$$\Rightarrow \frac{\partial}{\partial z} z^2 z^* = \frac{1}{2}\left(\frac{\partial}{\partial x} - j\frac{\partial}{\partial y}\right)\left[\left(x^3 + xy^2\right) + j\left(y^3 + x^2 y\right)\right]$$

$$= \frac{1}{2}\left\{\left[\left(3x^2 + y^2\right) + j2xy\right] - j2xy + \left(3y^2 + x^2\right)\right\}$$

$$= 2x^2 + 2y^2 = \underline{2|z|^2 = 2zz^*}$$

(b)

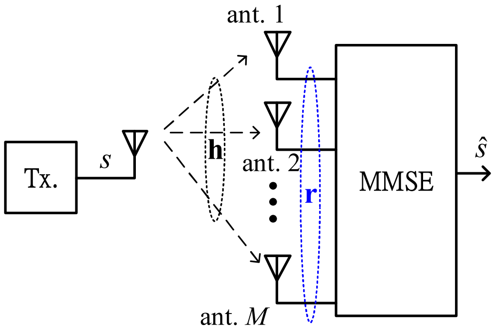
• $\mathbf{z}^T \mathbf{p} = \sum_{i=1}^{3} p_i z_i, \Rightarrow \frac{\partial}{\partial z_k} \mathbf{z}^T \mathbf{p} = \sum_{i=1}^{3} p_i \frac{\partial}{\partial z_k} z_i = p_k, k = 1, \cdots, 3$

$$\Rightarrow \frac{\partial}{\partial \mathbf{z}} \mathbf{z}^T \mathbf{p} = \begin{bmatrix} \sum_{i=1}^{n} p_i \frac{\partial}{\partial z_1} z_i \\ \vdots \\ \sum_{i=1}^{n} p_i \frac{\partial}{\partial z_n} z_i \end{bmatrix} = \begin{bmatrix} p_1 \\ \vdots \\ p_3 \end{bmatrix} = \mathbf{p} = \begin{bmatrix} 1+2i \\ 2+4i \\ 3 \end{bmatrix}$$

3. Consider a single input multiple output (SIMO) system equipping a single transmitting and M receiving antennas as shown in the figure below. The transmitted signal is s and the received signal vector is $\mathbf{r} = [r_1 \ \cdots \ r_M]^T = s\mathbf{h} + \mathbf{z}$, where $\mathbf{h} = [h_1 \ \cdots \ h_M]^T$ and $\mathbf{z} = [z_1 \ \cdots \ z_M]^T$ denote the channel response and additive white Gaussian noise vectors, respectively. The receiver uses a beamformer with a weight vector $\mathbf{w}_{\text{mmse}}$, which is determined under the minimum mean square error criterion. Given that $\mathrm{E}\{|s|^2\} = \sigma_s^2$, $\mathrm{E}\{\mathbf{z}\mathbf{z}^H\} = \sigma_z^2\mathbf{I}$, (20%)

(a) Please derive $\mathbf{w}_{\text{mmse}}$, when $\mathbf{h}$ or/and σ_z^2 is unknown to the receiver

(b) Please derive $\mathbf{w}_{\text{mmse}}$, when both $\mathbf{h}$ and σ_z^2 are known to the receiver



⟨Sol.⟩

(a)

• cost function:

$$C(\mathbf{w}) = E\left\{\left|s - \mathbf{w}^H\mathbf{r}\right|^2\right\}$$

$$= E\left\{\left(s - \mathbf{w}^H\mathbf{r}\right)\left(s^* - \mathbf{r}^H\mathbf{w}\right)\right\}$$

$$= \sigma_s^2 - \overbrace{E\left\{s\mathbf{r}^H\right\}}^{\mathbf{p}_{rs}^H}\mathbf{w} - \mathbf{w}^H \overbrace{E\left\{\mathbf{r}s^*\right\}}^{\mathbf{p}_{rs}} + \mathbf{w}^H \overbrace{E\left\{\mathbf{r}\mathbf{r}^H\right\}}^{\mathbf{R}_\mathbf{r}}\mathbf{w}$$

$$= \sigma_s^2 - \mathbf{p}_{rs}^H\mathbf{w} - \mathbf{w}^H\mathbf{p}_{rs} + \mathbf{w}^H\mathbf{R}_\mathbf{r}\mathbf{w}$$

• $MMSE$: $\left.\frac{\partial}{\partial\mathbf{w}}C(\mathbf{w})\right|_{\mathbf{w}_{\text{mmse}}} = \mathbf{0}, \Rightarrow \mathbf{R}_\mathbf{r}\mathbf{w}_{\text{mmse}} = \mathbf{p}_{rs}, \Rightarrow \mathbf{w}_{\text{mmse}} = \mathbf{R}_\mathbf{r}^{-1}\mathbf{p}_{rs}$

when $\mathbf{h}$ or/and σ_z^2 is unknown: $\begin{cases} \hat{\mathbf{R}}_\mathbf{r} = \dfrac{1}{N}\sum_{n=1}^{n}\mathbf{r}_n\mathbf{r}_n^H \\ \hat{\mathbf{p}}_{rs} = \dfrac{1}{N}\sum_{n=1}^{n}\mathbf{r}_n s_n^* \end{cases}$

(b)

• $\begin{cases} \mathbf{p}_{rs} = E\left\{\mathbf{r}s^*\right\} = E\left\{(s\mathbf{h} + \mathbf{z})s^*\right\} = \sigma_s^2\mathbf{h} \\ \mathbf{R}_\mathbf{r} = E\left\{\mathbf{r}\mathbf{r}^H\right\} = E\left\{(s\mathbf{h} + \mathbf{z})(s\mathbf{h} + \mathbf{z})^H\right\} = \sigma_s^2\mathbf{h}\mathbf{h}^H + \sigma_z^2\mathbf{I} \end{cases}$

$$\Rightarrow \mathbf{w}_{\text{mmse}} = \mathbf{R}_\mathbf{r}^{-1}\mathbf{p}_{rs} = \sigma_s^2\left(\sigma_s^2\mathbf{h}\mathbf{h}^H + \sigma_z^2\mathbf{I}\right)^{-1}\mathbf{h} = \left(\mathbf{h}\mathbf{h}^H + \overbrace{\frac{\sigma_z^2}{\sigma_s^2}}^{\rho}\mathbf{I}\right)^{-1}\mathbf{h} = \left(\mathbf{h}\mathbf{h}^H + \rho\mathbf{I}\right)^{-1}\mathbf{h}$$

where $\rho = \dfrac{\sigma_z^2}{\sigma_s^2} = SNR.$